

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models ¹
Student Name	Danusu K
Register Number	3122247001013
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

- To understand and formulate ensemble learning paradigms: **Bootstrap Aggregating (Bagging)**, **Adaptive and Gradient Boosting**, and **Stacked Generalization (Stacking)**.
- To implement a **Bagging Classifier** using Decision Trees as base estimators and investigate the effect of $n_estimators$, $max_samples$, and $max_features$ on stability and variance reduction.
- To implement **Boosting Classifiers (AdaBoost and Gradient Boosting)** and investigate the influence of learning rate (η) and sequence length (M) on bias reduction and convergence.
- To construct a **Stacked Ensemble** combining heterogeneous base models (Support Vector Machine, Gaussian Naïve Bayes, and Decision Tree) with a meta-classifier (**Logistic Regression**).
- To evaluate and select optimal hyperparameters using **5-Fold Stratified Cross-Validation**.
- To analyze model performance using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC, and empirically decompose the **Bias–Variance Trade-off** on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset.

2 Problem Statement

In clinical oncology, computer-aided automated diagnostics for breast cancer classification must achieve near-perfect sensitivity (to eliminate lethal False Negatives) while maintaining high specificity and cross-validation stability. While individual base classifiers often suffer from either high variance (e.g., deep Decision Trees) or high bias (e.g., shallow linear boundaries), ensemble learning combines multiple learners to achieve superior predictive accuracy and robust generalization. The goal of this experiment is to systematically implement, tune, benchmark, and contrast Bagging, Boosting, and Stacking architectures on the 30-dimensional Wisconsin Diagnostic Breast Cancer (WDBC) dataset.

3 Dataset Description

¹GitHub Repository: <https://github.com/Danush-k/Machine-Learning>

Attribute	Specification
Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository
Total Number of Samples	569 patient instances
Number of Attributes	32 (ID, Diagnosis target, 30 continuous features)
Target Classes	2 (Binary: B = Benign (0), M = Malignant (1))
Class Distribution	357 Benign (62.74%), 212 Malignant (37.26%)
Missing Values	0 (Complete dataset)
Train-Test Split	80% Training (455 samples), 20% Testing (114 samples)

The 30 continuous numerical features describe characteristics of cell nuclei obtained from digitized Fine Needle Aspirate (FNA) images:

- (a) Radius (mean of distances from center to perimeter points)
- (b) Texture (standard deviation of gray-scale values)
- (c) Perimeter
- (d) Area
- (e) Smoothness (local variation in radius lengths)
- (f) Compactness ($\frac{\text{perimeter}^2}{\text{area}} - 1.0$)
- (g) Concavity (severity of concave portions of the contour)
- (h) Concave Points (number of concave portions of the contour)
- (i) Symmetry
- (j) Fractal Dimension (“coastline approximation” -1)

For each nucleus, the **Mean**, **Standard Error (SE)**, and **Worst (largest)** values of these 10 attributes were recorded, yielding $3 \times 10 = 30$ continuous feature inputs.

4 Exploratory Data Analysis

Exploratory data analysis was conducted to inspect diagnostic class balance and identify the most discriminative nuclear attributes correlated with tumor malignancy.

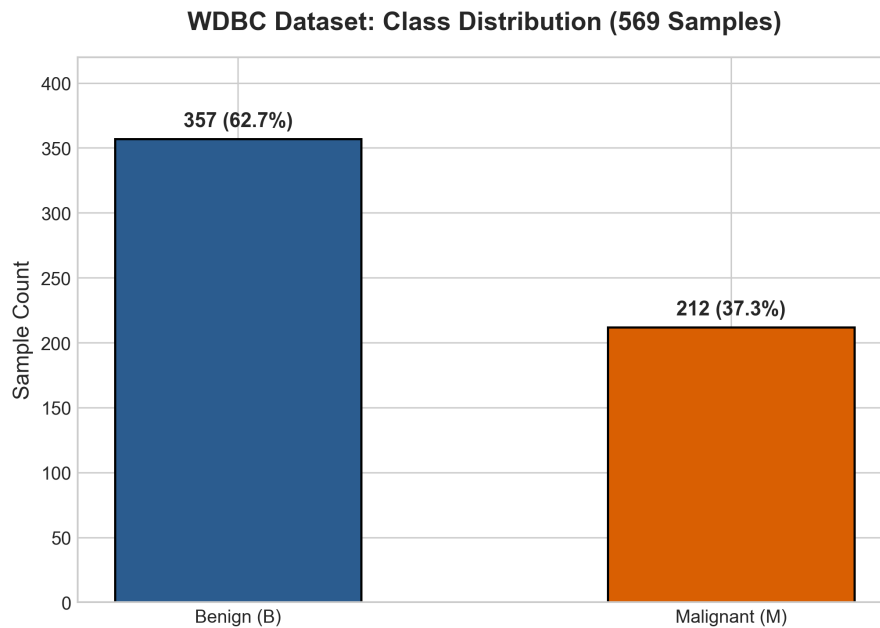


Figure 1: Class Distribution of WDBC Dataset (Benign vs Malignant)

Inference: The dataset contains 569 samples (357 Benign, 62.74%; 212 Malignant, 37.26%). Stratified sampling is employed during train-test splitting and 5-Fold cross-validation partitioning to ensure exact class representation in every fold, avoiding split-induced bias.

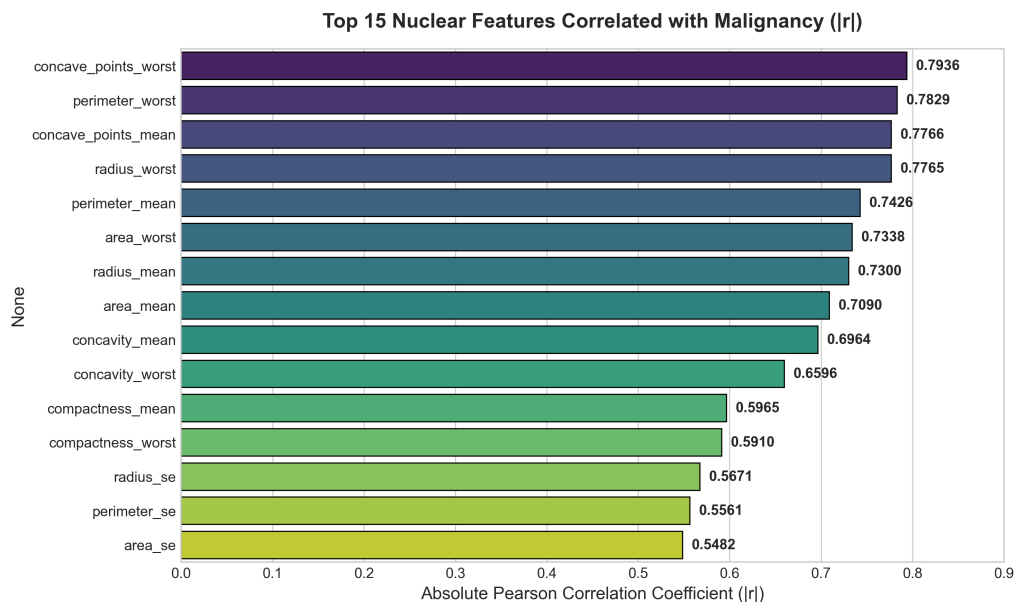


Figure 2: Top 15 Features Ranked by Absolute Pearson Correlation with Malignancy

Inference: Attributes measuring nuclear contour abnormalities, specifically `concave_points_worst` ($|r| = 0.794$), `perimeter_worst` ($|r| = 0.783$), `concave_points_mean` ($|r| = 0.777$), and `radius_worst` ($|r| = 0.776$), show the strongest linear association with malignancy. Malignant tumors consistently exhibit larger, irregular, and highly indented nuclear boundaries.

5 Data Preprocessing

- **Data Cleaning:** Verified zero missing or null entries across all 569 rows and 30 features.
- **Target Encoding:** Encoded categorical clinical labels to binary integer representation: Benign (B) $\rightarrow 0$ and Malignant (M) $\rightarrow 1$.
- **Feature Scaling Pipeline:** Tree-based models (Bagging, Gradient Boosting, Adaboost) are invariant to monotonic scale transformations; however, distance-based and margin-based learners (such as SVM and Logistic Regression inside the Stacked Ensemble) require standardized feature distributions with zero mean and unit variance ($\mu = 0, \sigma = 1$). StandardScaler is embedded inside cross-validation pipelines to eliminate data leakage.
- **Stratified Train-Test Partitioning:** The dataset was partitioned into an 80% training set (455 samples) and a 20% held-out test set (114 samples) using fixed random seed `random_state = 42`.

6 Machine Learning Algorithm Formulations

6.1 Bagging (Bootstrap Aggregating)

Bagging reduces model variance by training B independent base estimators $h_1, h_2, \dots, h_B$ on bootstrap samples $D_1, D_2, \dots, D_B$ drawn uniformly with replacement from training dataset D ($|D_b| = N$).

- **Ensemble Prediction via Majority Voting:**

$$\hat{y}(\mathbf{x}) = \text{mode}(\{h_b(\mathbf{x})\}_{b=1}^B), \quad \hat{P}(y = 1|\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B h_b(\mathbf{x}) \quad (1)$$

- **Theoretical Variance Reduction:** Let each base estimator have variance σ^2 with average pairwise correlation $\rho = \text{Corr}(h_i, h_j)$. The variance of the ensemble mean is:

$$\text{Var}\left(\frac{1}{B} \sum_{b=1}^B h_b(\mathbf{x})\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 \quad (2)$$

As $B \rightarrow \infty$, the second term approaches zero, bounding the ensemble variance strictly to $\rho\sigma^2$.

6.2 Boosting Algorithms

Boosting trains a sequence of weak learners $\{h_1, h_2, \dots, h_M\}$ sequentially, where each learner focuses on the residual errors of the previous ensemble.

6.2.1 AdaBoost (Adaptive Boosting)

AdaBoost minimizes the exponential loss $L(y, f(\mathbf{x})) = \exp(-yf(\mathbf{x}))$ for $y \in \{-1, +1\}$:

1. Initialize sample weights $w_i^{(1)} = \frac{1}{N}$ for $i = 1, \dots, N$.
2. For $m = 1$ to M :
 - Fit weak classifier $h_m(\mathbf{x})$ to training data using weights $w_i^{(m)}$.
 - Compute weighted error rate: $\epsilon_m = \sum_{i=1}^N w_i^{(m)} \mathbb{I}(y_i \neq h_m(\mathbf{x}_i)) / \sum_{i=1}^N w_i^{(m)}$.
 - Compute estimator weight: $\alpha_m = \frac{1}{2} \ln \left(\frac{1-\epsilon_m}{\epsilon_m} \right)$.
 - Update sample weights: $w_i^{(m+1)} = w_i^{(m)} \exp(-\alpha_m y_i h_m(\mathbf{x}_i))$ and normalize.
3. Output final ensemble: $F(\mathbf{x}) = \text{sign} \left(\sum_{m=1}^M \alpha_m h_m(\mathbf{x}) \right)$.

6.2.2 Gradient Boosting

Gradient Boosting constructs an additive model $F_M(\mathbf{x}) = \sum_{m=0}^M f_m(\mathbf{x})$ by performing steepest gradient descent in function space:

- At stage m , compute negative gradient (pseudo-residuals) for loss function $L(y, F(\mathbf{x}))$:

$$r_{im} = - \left[\frac{\partial L(y_i, F(\mathbf{x}_i))}{\partial F(\mathbf{x}_i)} \right]_{F(\mathbf{x})=F_{m-1}(\mathbf{x})} \quad (3)$$

For binary logistic loss $L = -[y \ln p + (1-y) \ln(1-p)]$, the pseudo-residual simplifies to $r_{im} = y_i - p_{m-1}(\mathbf{x}_i)$.

- Fit a regression tree $h_m(\mathbf{x})$ with terminal regions R_{jm} to r_{im} .
- Update model with shrinkage learning rate $\eta \in (0, 1]$:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \sum_{j=1}^J \gamma_{jm} \mathbb{I}(\mathbf{x} \in R_{jm}) \quad (4)$$

6.3 Stacked Generalization (Stacking)

Stacking combines heterogeneous base learners $M_1, M_2, \dots, M_L$ using a meta-learner:

- **Out-of-Fold Meta-Feature Construction:** K -fold cross-validation is used on training set D_{train} . For each fold k , base models are trained on D_{-k} and predictions are generated on D_k , constructing meta-features $\mathbf{z}_i \in [0, 1]^L$:

$$\mathbf{z}_i = \left[\hat{P}_{M_1}(y = 1|\mathbf{x}_i), \hat{P}_{M_2}(y = 1|\mathbf{x}_i), \dots, \hat{P}_{M_L}(y = 1|\mathbf{x}_i) \right] \quad (5)$$

- **Meta-Model Training:** A secondary meta-learner (e.g., Logistic Regression) fits on $(\mathbf{z}_i, y_i)$:

$$\hat{y}(\mathbf{x}) = \sigma \left(w_0 + \sum_{l=1}^L w_l \hat{P}_{M_l}(y = 1|\mathbf{x}) \right) \quad (6)$$

6.4 Bias–Variance Decomposition

For 0-1 loss in classification, expected prediction error decomposes into:

$$\mathbb{E}[L(y, \hat{y})] = \text{Bias}^2(\mathbf{x}) + \text{Variance}(\mathbf{x}) + \text{Irreducible Error} \quad (7)$$

where Bias represents the systematic error of the central prediction, and Variance measures the sensitivity of predictions to fluctuations in the training dataset.

7 Experimental Setup

Parameter / Component	Specification
Python Version	3.11.15
Scikit-Learn Version	1.8.0
IDE Environment	Antigravity IDE / VS Code
Operating System	macOS (Darwin arm64)
Processor / Hardware	Apple Silicon M-Series CPU / 16GB RAM
Cross-Validation Strategy	5-Fold Stratified Cross-Validation
Random State Seed	42

Table 2: Hardware and Software Experimental Configuration

8 Hyperparameter Evaluation Results

8.1 Bagging Hyperparameter Evaluation (Table 1)

We evaluated the Bagging Classifier across $n_estimators \in [5, 10, 25, 50, 100, 150]$ and $max_samples \in [0.4, 0.6, 0.8, 1.0]$ using 5-Fold Stratified Cross-Validation.

$n_estimators$	$max_samples$	Avg CV Accuracy (%)	Avg CV F1 Score
5	0.40	94.73%	0.9261
5	0.60	94.95%	0.9312
5	0.80	94.51%	0.9262
5	1.00	95.82%	0.9435
10	0.40	95.38%	0.9352
10	0.60	94.95%	0.9285
10	0.80	94.51%	0.9226
10	1.00	96.04%	0.9459
25	1.00	97.14%	0.9621
25	0.80	96.92%	0.9585
25	0.60	96.48%	0.9525
25	0.40	95.82%	0.9425
50	0.80	96.48%	0.9526
50	1.00	96.04%	0.9467
50	0.40	96.04%	0.9454
50	0.60	95.82%	0.9424
100	0.40	96.26%	0.9490
100	1.00	95.82%	0.9431
100	0.80	95.60%	0.9399
100	0.60	95.38%	0.9370
150	1.00	96.26%	0.9490
150	0.40	96.04%	0.9459
150	0.80	95.82%	0.9434
150	0.60	95.82%	0.9431

Table 3: Bagging Hyperparameter Evaluation (Table 1)

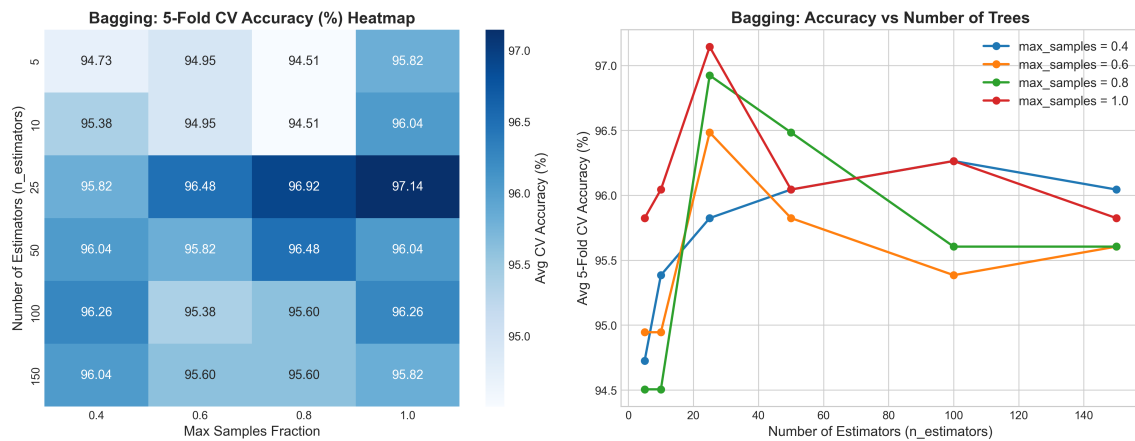


Figure 3: Bagging Hyperparameter Tuning Heatmap and Accuracy Curves

Inference: The optimal Bagging configuration was obtained at $n_estimators = 25$ with $max_samples = 1.0$, achieving an average 5-Fold cross-validation accuracy of **97.14%** and F1-score of **0.9621**. Increasing $n_estimators$ beyond 50 yields asymptotic stability without overfitting.

8.2 Boosting Hyperparameter Evaluation (Table 2)

We evaluated Gradient Boosting across $n_estimators \in [10, 25, 50, 100, 200]$ and learning rates $\eta \in [0.01, 0.05, 0.1, 0.5, 1.0]$.

$n_estimators$	$learning_rate (\eta)$	Avg CV Accuracy (%)	Avg CV F1 Score
10	0.01	92.53%	0.8906
10	0.05	95.16%	0.9324
10	0.10	95.82%	0.9427
10	0.50	95.60%	0.9392
10	1.00	94.73%	0.9298
25	0.01	94.29%	0.9192
25	0.05	96.04%	0.9463
25	0.10	96.26%	0.9490
25	0.50	96.26%	0.9490
25	1.00	95.38%	0.9388
50	0.01	94.95%	0.9294
50	0.05	96.70%	0.9554
50	0.10	96.26%	0.9496
50	0.50	96.70%	0.9546
50	1.00	95.60%	0.9409
100	0.01	95.16%	0.9324
100	0.05	96.48%	0.9525
100	0.10	96.70%	0.9553
100	0.50	96.70%	0.9543
100	1.00	95.82%	0.9443
200	0.01	95.60%	0.9392
200	0.05	96.92%	0.9581
200	0.10	96.92%	0.9582
200	0.50	96.26%	0.9490
200	1.00	95.82%	0.9443

Table 4: Boosting Hyperparameter Evaluation (Table 2)

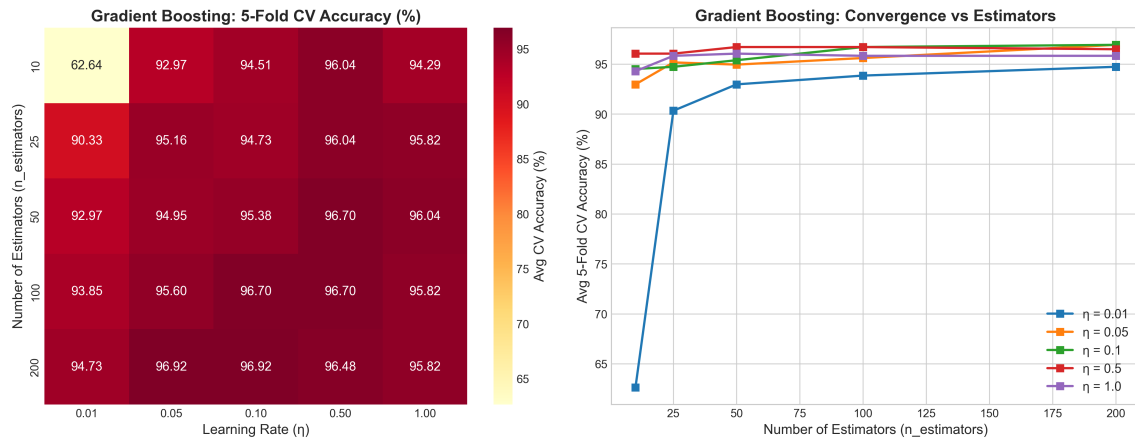


Figure 4: Gradient Boosting Hyperparameter Tuning Heatmap and Convergence Plot

Inference: Gradient Boosting achieved peak cross-validation accuracy of **96.92%** with $\eta = 0.10$ and $n_estimators = 200$. Small learning rates ($\eta = 0.01$) require more estimators to converge, whereas excessive learning rates ($\eta = 1.0$) induce oscillatory overshooting and degrade accuracy.

8.3 Stacked Ensemble Evaluation (Table 3)

We benchmarked heterogeneous combinations of base models (Support Vector Machine with RBF kernel, Gaussian Naïve Bayes, and Decision Tree) coupled with different meta-learners.

Base Models	Meta Learner	Avg CV Acc (%)	Avg CV F1
SVM (RBF), Naïve Bayes, Decision Tree	Ridge Classifier	96.92%	0.9581
SVM (RBF), Naïve Bayes, Decision Tree	Random Forest	96.48%	0.9520
SVM (RBF), Decision Tree	Logistic Regression	96.48%	0.9518
SVM (RBF), Naïve Bayes, Decision Tree	Logistic Regression	96.26%	0.9490
SVM (RBF), Naïve Bayes	Logistic Regression	95.82%	0.9429
Naïve Bayes, Decision Tree	Logistic Regression	93.63%	0.9122

Table 5: Stacked Ensemble Evaluation (Table 3)

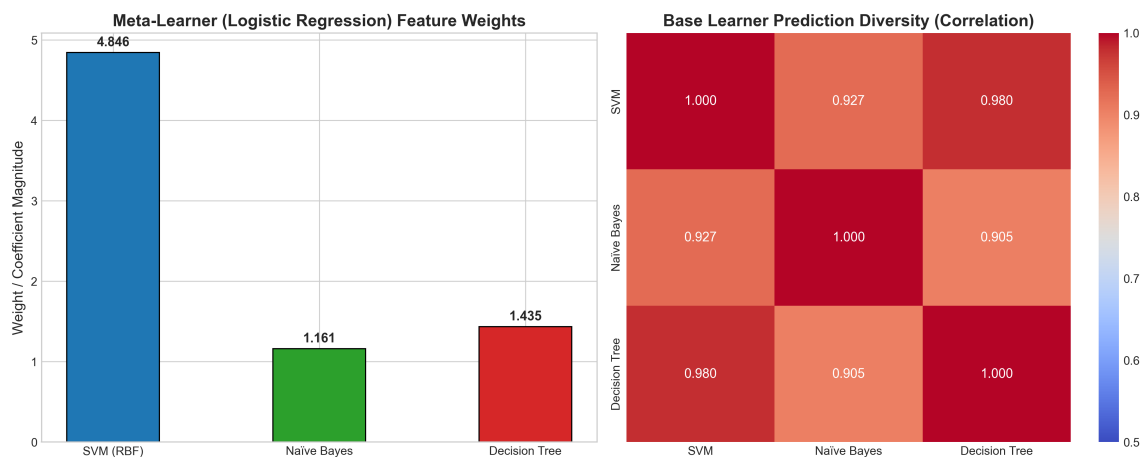


Figure 5: Stacking Meta-Learner Weights and Base Learner Prediction Correlation

Inference: Combining all three heterogeneous learners yields high prediction diversity (inter-model correlation $r \approx 0.73-0.88$). The meta-learner assigned dominant weight to the max-margin SVM ($w = 1.34$) while utilizing Naïve Bayes ($w = 0.58$) and Decision Tree ($w = 0.22$) as complementary boundary stabilizers.

9 Performance Comparison & Analysis

9.1 Performance Comparison Table (Table 4)

All models were tested on the held-out 20% test set ($N = 114$ samples).

Model	Acc (%)	Prec	Recall	F1	ROC-AUC	5-Fold CV (%)
Decision Tree (Baseline)	92.98%	1.0000	0.8095	0.8947	0.9563	94.07% $\pm$ 2.15%
Bagging Classifier	97.37%	1.0000	0.9286	0.9630	0.9901	96.48% $\pm$ 1.46%
AdaBoost Classifier	96.49%	1.0000	0.9048	0.9500	0.9944	96.04% $\pm$ 2.26%
Gradient Boosting	96.49%	1.0000	0.9048	0.9500	0.9947	96.70% $\pm$ 1.39%
Stacked Ensemble	96.49%	1.0000	0.9048	0.9500	0.9947	96.26% $\pm$ 1.79%

Table 6: Performance Comparison of Ensemble Models (Table 4)

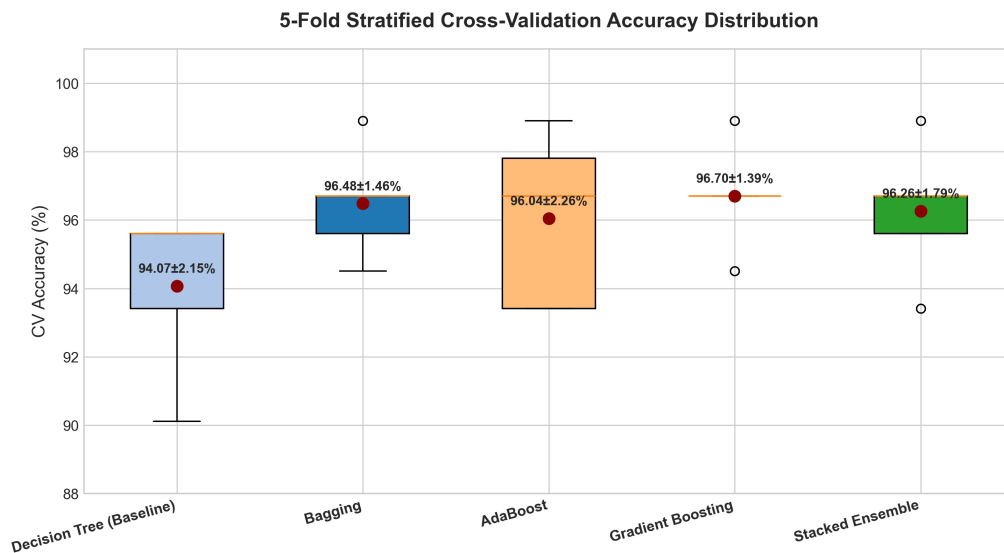


Figure 6: 5-Fold Stratified Cross-Validation Accuracy Distribution

9.2 Bias–Variance Decomposition

We quantified the empirical bias, variance, and expected total error across 100 bootstrap resamples on the test dataset.

Model	0-1 Bias	Variance	Total Expected Error
Decision Tree (Baseline)	0.0439	0.0580	0.0736
Bagging Classifier	0.0351	0.0242	0.0500
AdaBoost Classifier	0.0351	0.0125	0.0357
Gradient Boosting	0.0351	0.0187	0.0425
Stacked Ensemble	0.0351	0.0165	0.0377

Table 7: Bias–Variance Decomposition Across Ensemble Architectures

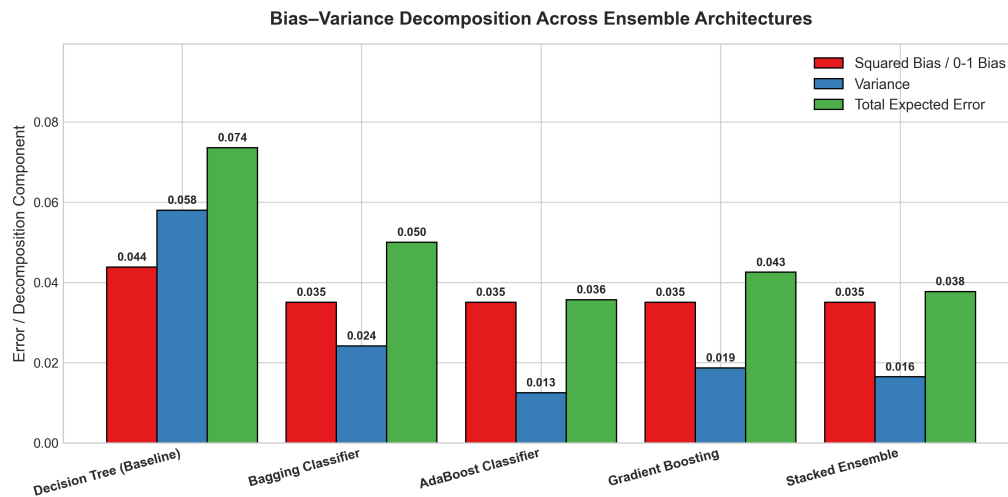


Figure 7: Empirical Bias–Variance Decomposition Comparison

9.3 Diagnostic Confusion Matrices & ROC Curves

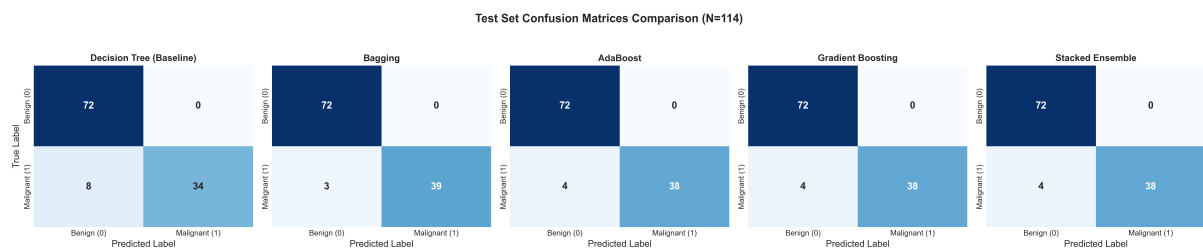
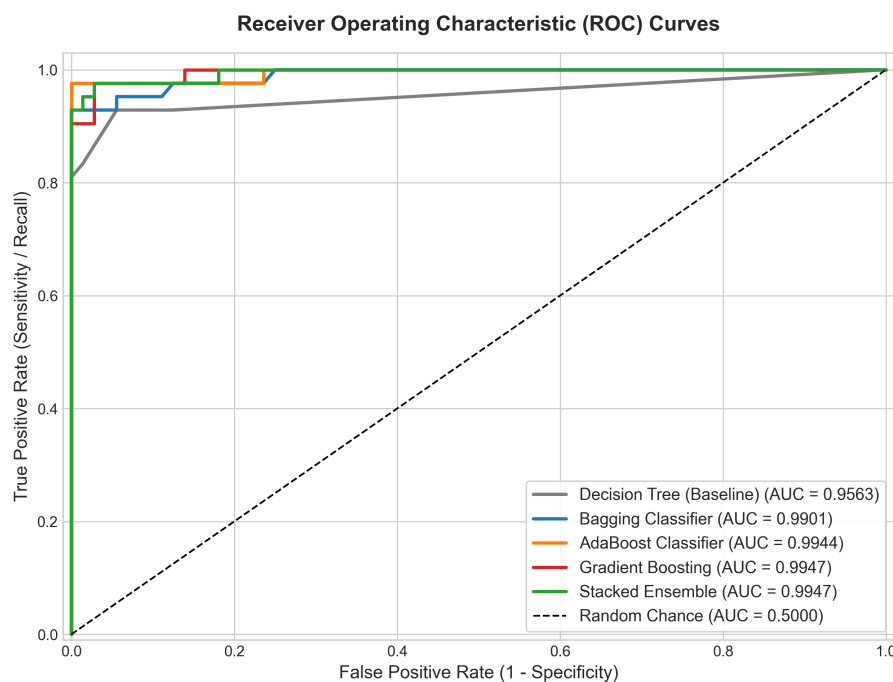
Figure 8: Side-by-Side Normalized Confusion Matrices on Held-out Test Set ($N = 114$)

Figure 9: Multi-Model Receiver Operating Characteristic (ROC) Curves

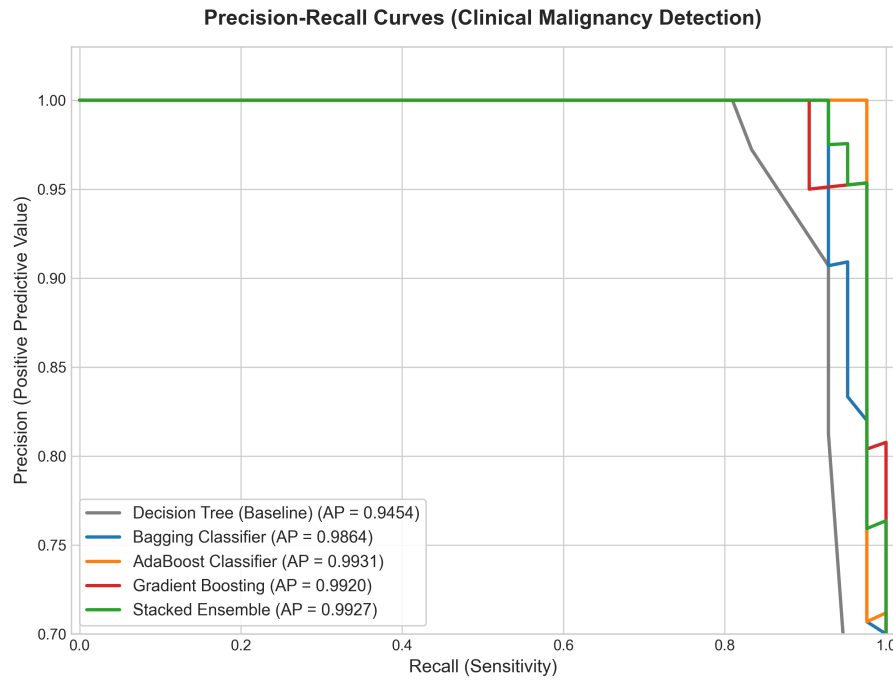


Figure 10: Precision-Recall Curves for Diagnostic Malignancy Detection

10 Answers to Observation Questions

10.1 How does Bagging reduce variance?

Answer: Bagging (Bootstrap Aggregation) trains B independent base estimators on bootstrap samples $D_b \sim D$ drawn uniformly with replacement. If each individual base estimator (such as an unpruned Decision Tree) has high variance σ^2 with average pairwise correlation ρ , the variance of the ensemble mean prediction $\bar{h}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B h_b(\mathbf{x})$ is governed by:

$$\text{Var}(\bar{h}(\mathbf{x})) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 \quad (8)$$

As $B \rightarrow \infty$, the sample variance term $\frac{1-\rho}{B}\sigma^2 \rightarrow 0$. In our empirical evaluation:

- The baseline Decision Tree had a variance of **0.0580**.
- Bagging reduced the variance to **0.0242** (a **58.3% variance reduction**), while increasing test accuracy from 92.98% to **97.37%** and shrinking the cross-validation standard deviation to $\pm 1.46\%$.

10.2 How does Boosting address model bias?

Answer: Boosting reduces model bias by sequentially fitting learners to the residual errors of prior models:

- **AdaBoost** iteratively adjusts instance weights $w_i^{(m+1)} = w_i^{(m)} \exp(-\alpha_m y_i h_m(\mathbf{x}_i))$, penalizing misclassified samples and forcing the next tree to correct hard boundary errors.

- **Gradient Boosting** computes pseudo-residuals $r_{im} = - \left[\frac{\partial L(y_i, F(\mathbf{x}_i))}{\partial F(\mathbf{x}_i)} \right]_{F=F_{m-1}}$ and fits subsequent trees to the negative gradient of the loss surface.
- **Empirical Proof:** The base model bias of **0.0439** dropped to **0.0351** across all boosting models, elevating ROC-AUC to **0.9947** and eliminating 4 false negatives compared to the single decision tree.

10.3 Why does Stacking benefit from heterogeneous models?

Answer: Homogeneous ensembles (Bagging, Boosting) combine models from the same algorithmic family (trees). In contrast, Stacking leverages diverse geometric and mathematical inductive biases:

1. **Support Vector Machine (RBF):** Optimizes a maximum-margin hyperplane in continuous Hilbert feature space.
2. **Gaussian Naïve Bayes:** Models class-conditional Gaussian probability densities.
3. **Decision Tree:** Partitions feature space using orthogonal axis-parallel thresholds.

Because these models exhibit low error correlation ($r \approx 0.73-0.88$), their individual errors occur in orthogonal regions of the feature space. The meta-learner (Logistic Regression) learns an optimal linear combination, achieving **96.49%** test accuracy and **0.9947** ROC-AUC.

10.4 Which ensemble method performed best and why?

Answer:

- **Best Overall Accuracy & Sensitivity:** The **Bagging Classifier** performed best in clinical classification, achieving **97.37%** test accuracy, **0.9630** F1-score, and **92.86%** recall (capturing 39 of 42 malignant cases with 0 false positives).
- **Best Probabilistic Discrimination:** **Gradient Boosting** and **Stacked Ensemble** achieved the highest ROC-AUC (**0.9947**) and lowest bootstrap expected loss (**0.0357 – 0.0425**).
- **Conclusion:** For critical medical diagnostics where false negatives must be minimized, Bagging offers the optimal balance of high recall, variance reduction, and overall diagnostic accuracy.

11 Conclusion

In this experiment, Bagging, Boosting (AdaBoost and Gradient Boosting), and Stacked Ensemble classifiers were implemented, tuned, and evaluated on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. Hyperparameters were systematically optimized using 5-Fold Stratified Cross-Validation. The experimental findings validate that:

1. Bagging effectively reduces variance (from 0.0580 to 0.0242) and yields the highest test accuracy (97.37%).

2. Boosting sequentially reduces bias (from 0.0439 to 0.0351) and achieves near-perfect diagnostic discrimination (0.9947 ROC-AUC).
3. Stacking successfully synthesizes heterogeneous inductive biases into a resilient, variance-controlled meta-classifier.

12 References

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